AI ACADEMY · PYTHON READINESS BRIDGE · STUDENT LESSON

Reading structured data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

Before you start

Which key stores the occupied count in {"capacity": 8, "used": 3}?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Process a list of records consistently. Try the recall before opening notes.

Your learning target

I can **process** a list of records consistently.

I can compute a total from named **field**s.

I can preserve record meaning while tracing a loop.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Dictionaries and records. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Functions and arguments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Process a list of records consistently

Compute a total from named **field**s

Preserve record meaning while tracing a loop

Retrieve what you already know

> Which key stores the occupied count in {"capacity": 8, "used": 3}?

My starting explanation

This week's end goal

Trace reading structured data and explain the changed program before running it.

Notice the evidence

> Investigate the supplied Reading structured data case. Separate observations, given facts and assumptions before choosing an explanation.

Two fictional cupboards are stored as a list of dictionaries. Each record uses the same keys and units. We want the total spare spaces. Treat the records as a **dataset**, and trace one record at a time before trusting a total.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Reading structured data. Connect each labelled input, deciding step and result.

```
rows = [{"capacity": 8, "used": 3},
        {"capacity": 6, "used": 6}]
total = 0
for row in rows:
    spare = row["capacity"] - row["used"]
    total = total + spare
print(total)
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
First row	8 - 3 = 5	total becomes 5
Second row	6 - 6 = 0	total stays 5

Read each row from left to right and connect the evidence to the stated rule.

The loop variable row refers to each record in turn. We calculate spare from that record, then add it. total starts before the loop.

An empty list produces 0 because there are no updates; it does not tell us whether no cupboards exist or records were not collected.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Reading structured data. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For spare contributions 2, 4 and 1, trace total starting at 0.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Add {"count":4} as a fourth row.

Predicted result

Prediction: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

Why the result follows

Reason: Only the added recorded value contributes; the existing absent **field** remains skipped.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Reading structured data to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Dataset	A collection of records
Row	One record in a table
Field	One named part of a record
Accumulator	A variable updated to keep a running result

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Reading structured data, predict a step, then compare with its explanation.

Add a third record with capacity 10 and used 4. Predict the total and list each contribution.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Reading structured data

Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

The facts we will use

Supplied facts

Python 3 records: [{"count":2},{"count":0},{}]. Policy: include existing numeric counts, including zero; skip an absent **field**.

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Included values and absent field

	Count value	Include in mean?
Row 1	2	Yes
Row 2	0	Yes
Row 3	Absent	No

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

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Reading structured data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect three rows

Two rows contain count; the third does not.



2. Accumulate values

The included values sum to 2+0=2, with denominator 2.



3. Report coverage and mean

Mean is 1; measured coverage is 2 of 3 rows.

CHANGE ONE THING

Add {"count":4} as a fourth row.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Inspect three rows	Two rows contain count; the third does not.
2. Accumulate values	The included values sum to 2+0=2, with denominator 2.
3. Report coverage and mean	Mean is 1; measured coverage is 2 of 3 rows.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Structured data need a row policy before aggregation. Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

What this example does not establish

Keep the stated limit

This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

Keep the stated limit

Read the labelled picture: Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

Keep the stated limit

Change one thing: Add {"count":4} as a fourth row. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Reading structured data claim and explain a repair using the supplied evidence.

Putting total = 0 inside the loop discards earlier contributions. With the original rows it finishes at 0, the second contribution. Move the initialization before the loop and rerun both the original and reversed row order.

Locate the earliest unsupported step

> A loop resets total to 0 on each pass over contributions 2, 4, 1. What final total appears and how should it be fixed?

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Reading structured data with support, then explain the deciding step in your own words.

In the worked **dataset**, state the number of records and the spare spaces in the second record.

Add a third record with capacity 10 and used 4. Predict the total and list each contribution.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace all updates

Use workbook W2 for the connected example “Reading structured data”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Reading structured data. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Reading structured data. Record the prediction, actual observation or labelled

> paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

For spare contributions 2, 4 and 1, trace total starting at 0.

```
rows = [{"capacity": 8, "used": 3},
        {"capacity": 6, "used": 6}]
total = 0
for row in rows:
    spare = row["capacity"] - row["used"]
    total = total + spare
print(total)
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Reading structured data. Explain why each step is justified.

Explain → apply → challenge

In the worked **dataset**, state the number of records and the spare spaces in the second record.

> A loop resets total to 0 on each pass over contributions 2, 4, 1. What final total appears and how should it be fixed?

> An empty **dataset** produces total 0. Can you conclude that every cupboard is full?

> **Explain which evidence supports your conclusion and which claim remains unproved.**

Connect your project

> Apply this week's idea to Python readiness portfolio. Keep an annotated decision and the evidence supporting it.

Add a labelled reading structured data **evidence** sheet to Python readiness portfolio. Show how you can **process** a list of records consistently.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

This loop assumes every record has the required integer **field**s. A missing used **field** causes KeyError. Do not silently treat a missing count as zero; that would invent unused capacity.

An empty dataset produces total 0. Can you conclude that every cupboard is full?

An empty **dataset** produces total 0. Can you conclude that every cupboard is full?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Reading structured data case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
First row	8 - 3 = 5	total becomes 5
Second row	6 - 6 = 0	total stays 5

Read each row from left to right and connect the evidence to the stated rule.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Preserve records and report measured coverage

Trace the named **field**s and missing-value policy. Predict the **output**, state the observed total and mean denominator, and report coverage. What changes if all three rows are empty records?

```
rows = [{"score": 2}, {"score": 6}, {}]
values = []
for row in rows:
    value = row.get("score")
    if value is not None:
        values.append(value)
observed_total = sum(values)
if len(values) > 0:
    mean = observed_total / len(values)
else:
```

```
    mean = None
print(observed_total, len(values), len(rows), mean)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Write a loop for records (capacity 9, used 2), (capacity 5, used 1), and (capacity 4, used 4). Predict the total. Reverse the order and test an empty list.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Reading structured data, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

By the end, I can	How I will show it
Process a list of records consistently	Explain the deciding step, show a trace or test, and state one limit.
Compute a total from named fields	Explain the deciding step, show a trace or test, and state one limit.
Preserve record meaning while tracing a loop	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

Finish and return

> Explain Reading structured data without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: **Process** a list of records consistently

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Mean is 1; measured coverage is 2 of 3 rows.

1. Inspect three rows

2. Report coverage and mean

3. Accumulate values

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report coverage and mean Mean is 1; measured coverage is 2 of 3 rows. Structured data need a row policy before aggregation.

Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add {"count":4} as a fourth row. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4. Only the added recorded value contributes; the existing absent **field** remains skipped.

Repair a missing building block

Review Dictionaries and records (python-bridge/10). Goal: Read and update named dictionary **field**s. Short practice: Predict all three original **output**s.

Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.. Return to this week and explain what you changed.

Review Functions and arguments (python-bridge/8). Goal: Distinguish a function definition from a function call. Short practice: Match every argument to a parameter for the two calls.

Why do **output**s differ, and what prints when both calls are removed?. Return to this week and explain what you changed.

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches.

Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

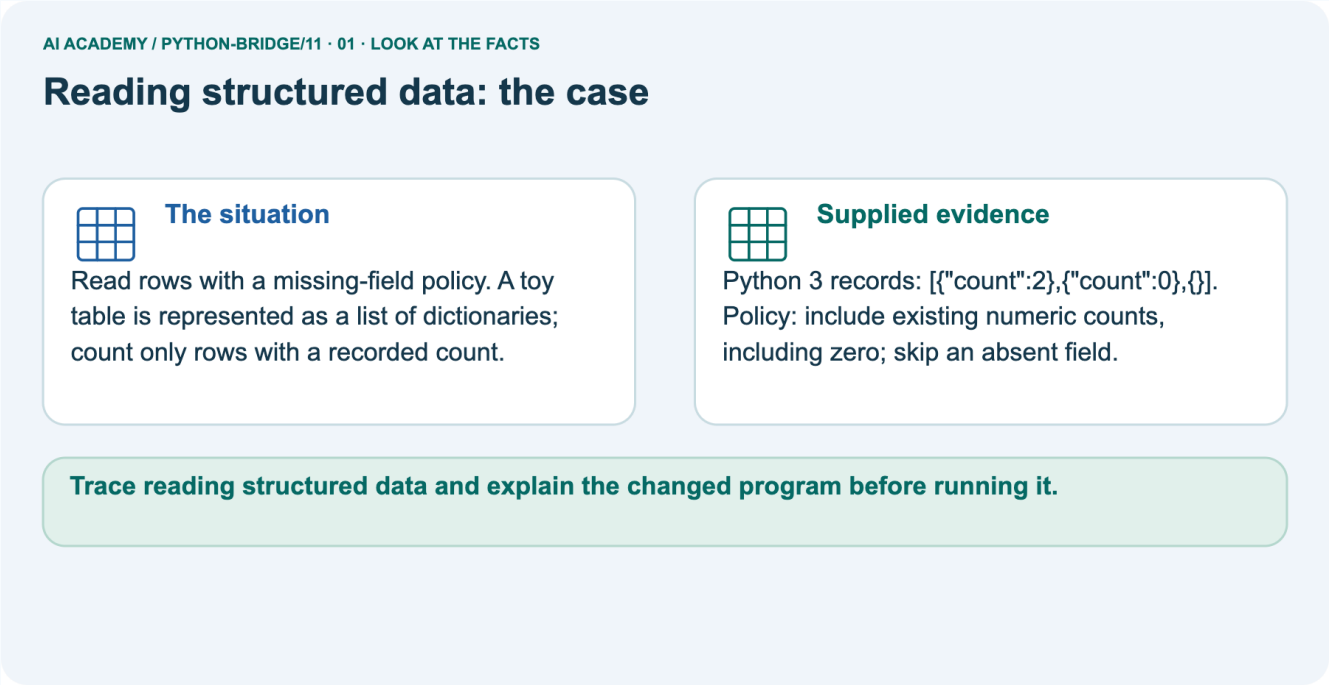
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Reading structured data graphic using its labels, deciding evidence and relevant limitation.

01. Reading structured data: the case



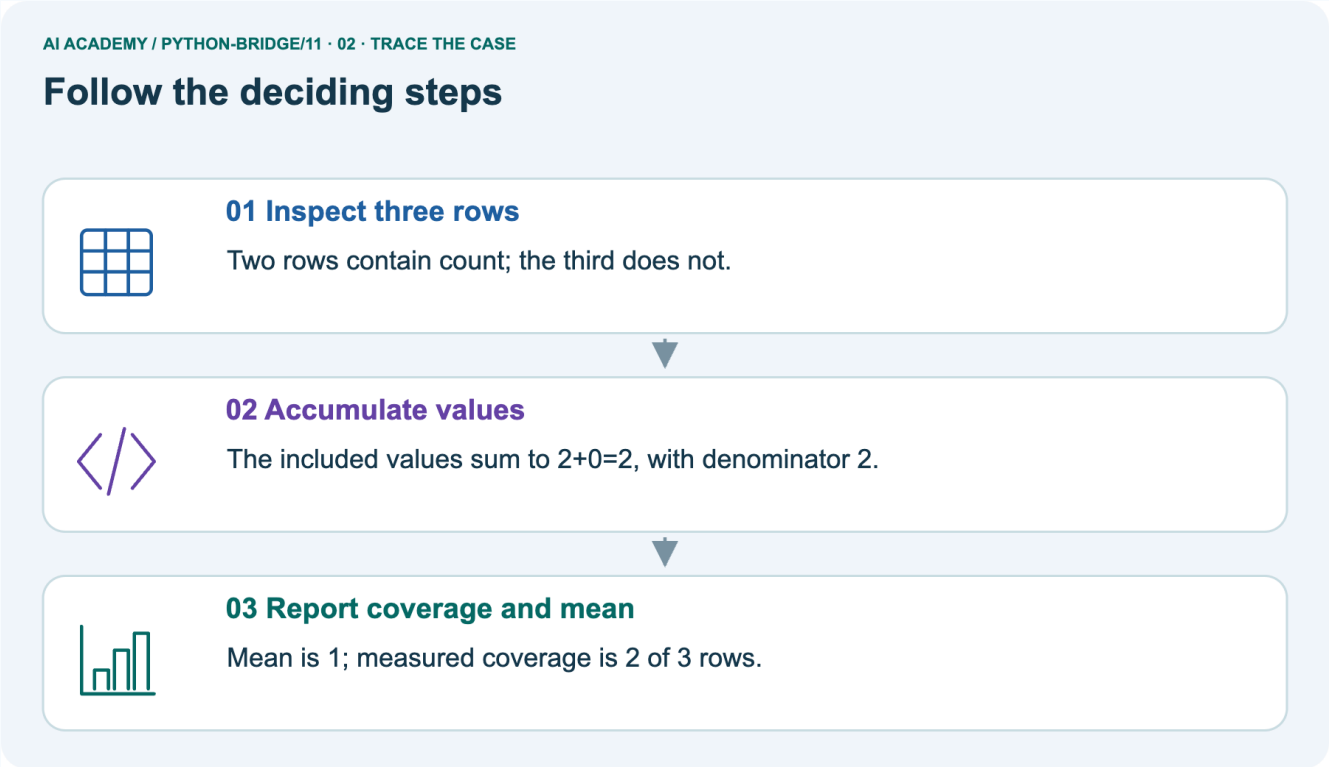
Follow the labelled picture with the matching explanation. Open the vector diagram

Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count. Supplied facts: Python 3 records: [{"count":2},{"count":0},{}].

Policy: include existing numeric counts, including zero; skip an absent **field**.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

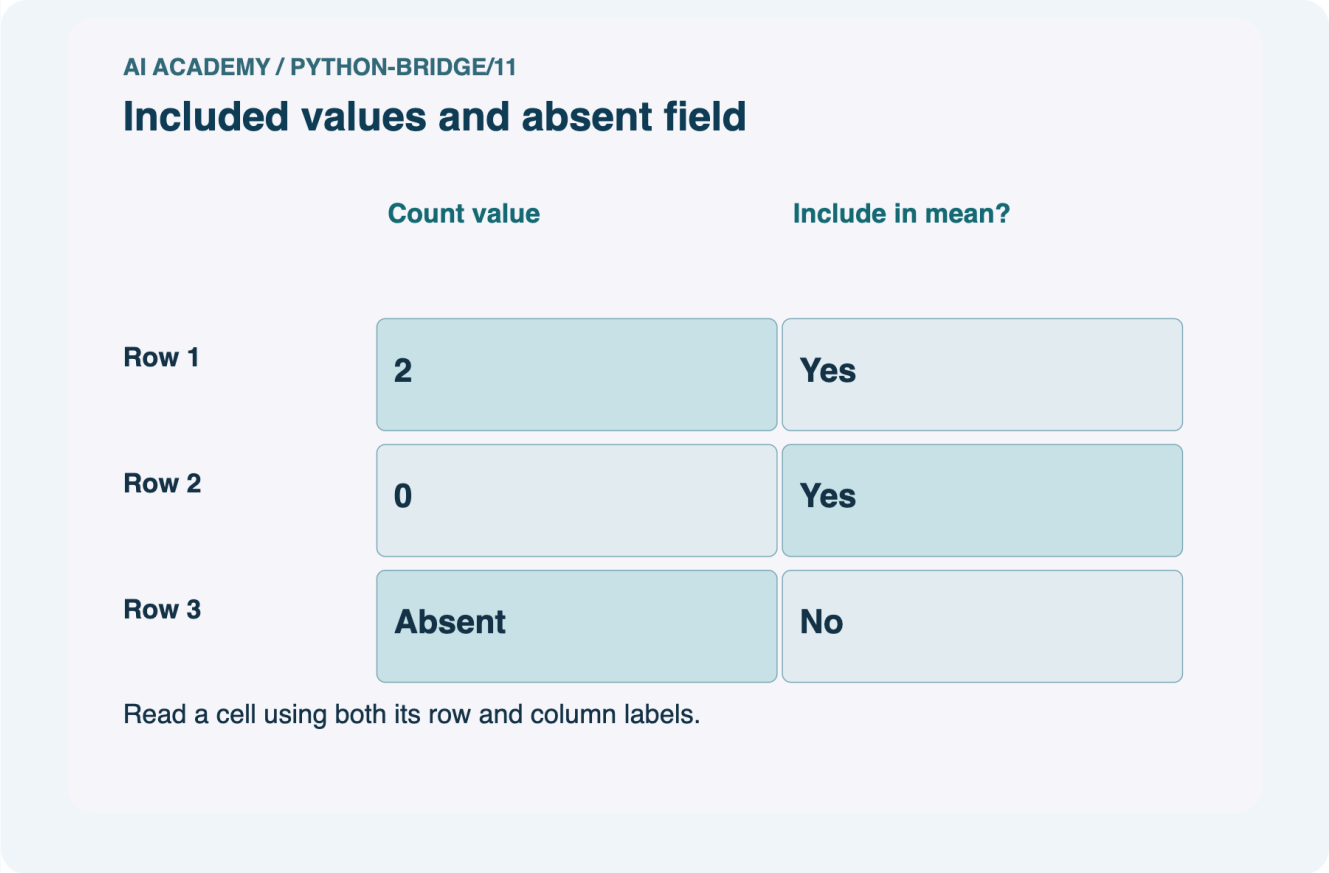


Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect three rows: Two rows contain count; the third does not. → Accumulate values: The included values sum to 2+0=2, with denominator 2. → Report coverage and mean: Mean is 1; measured coverage is 2 of 3 rows.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Included values and absent field

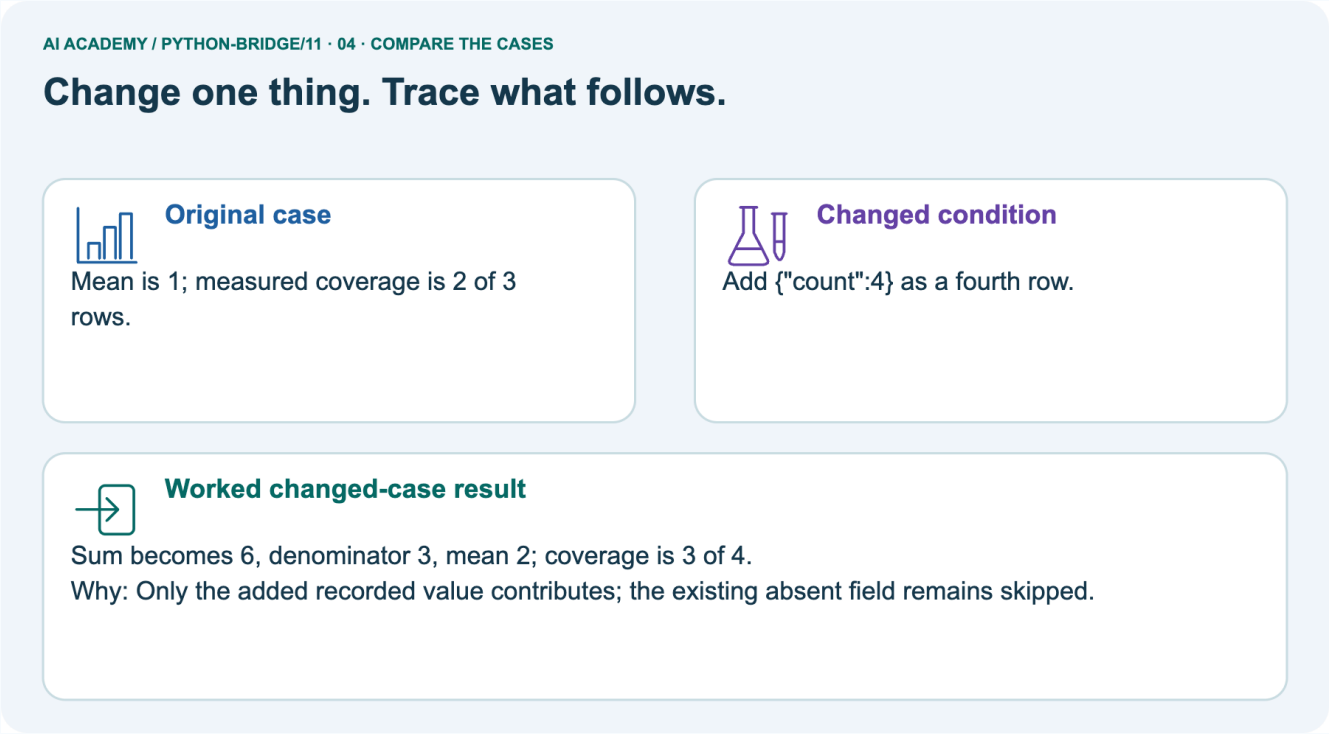


Follow the labelled picture with the matching explanation. Open the vector diagram

Included values and absent **field**. Row 1: Count value = 2, Include in mean? = Yes. Row 2: Count value = 0, Include in mean? = Yes. Row 3: Count value = Absent, Include in mean? = No.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



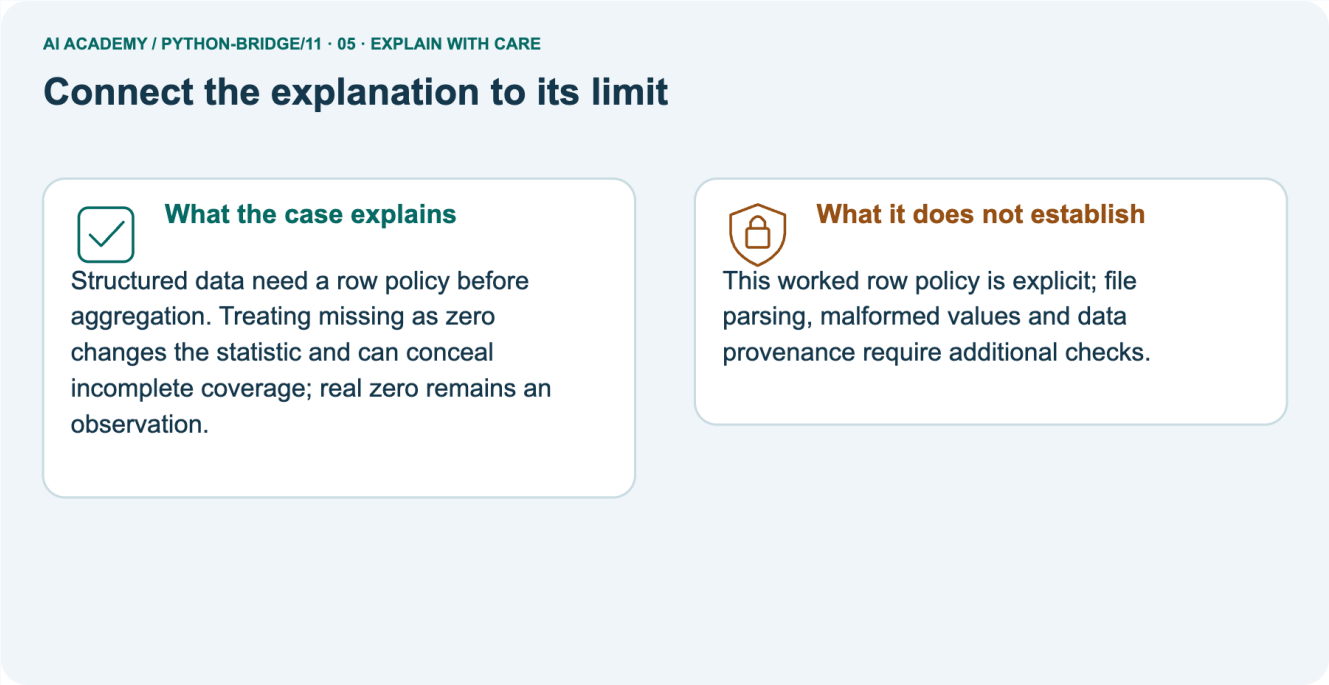
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Mean is 1; measured coverage is 2 of 3 rows. Changed condition: Add {"count":4} as a fourth row. Expected result: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

Reason: Only the added recorded value contributes; the existing absent **field** remains skipped.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



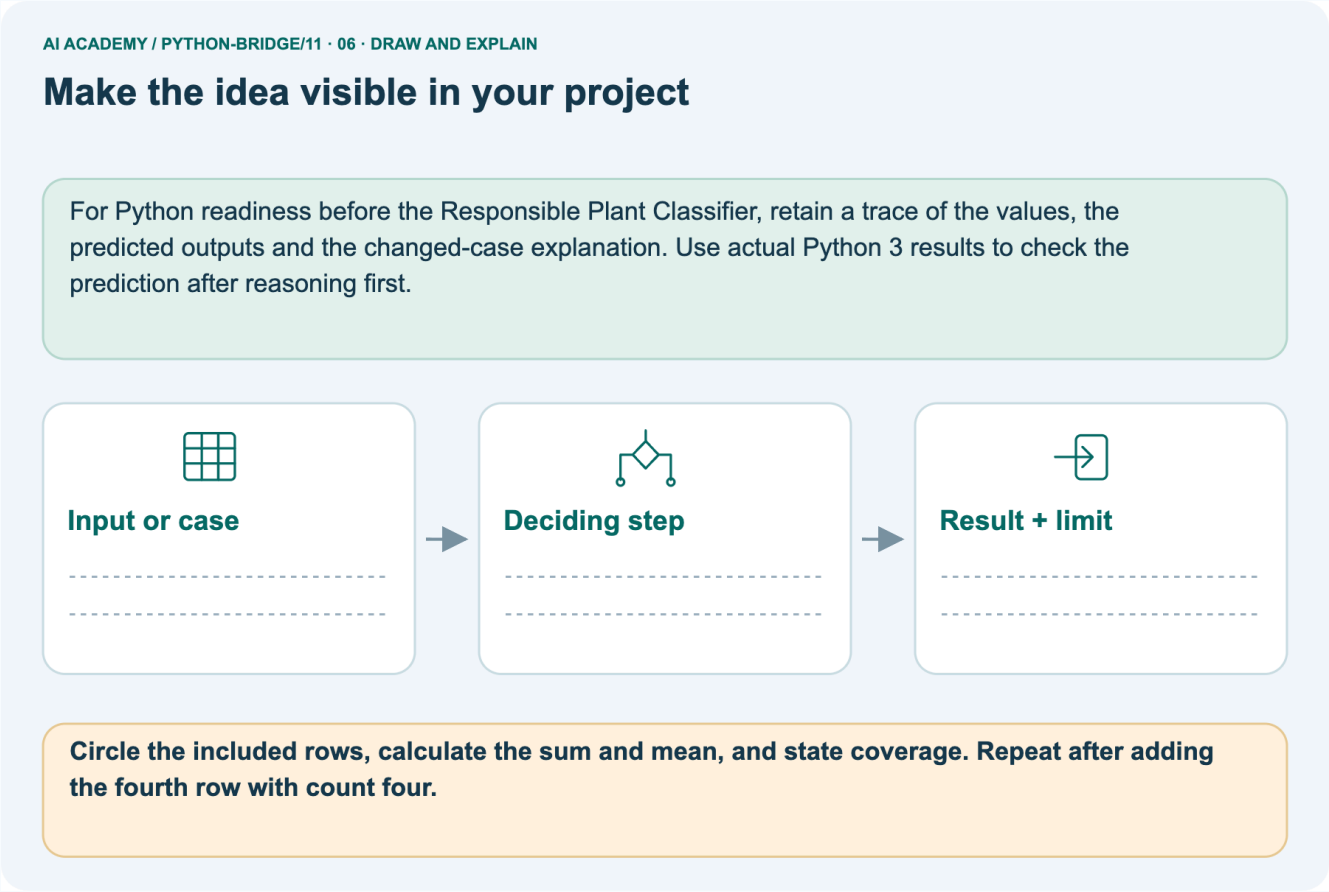
Follow the labelled picture with the matching explanation. Open the vector diagram

Structured data need a row policy before aggregation. Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

Limit: This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.